CHAPTER 28: TRAFFIC SIGNAL MAINTENANCE

28.1 INTRODUCTION

1 Adequate maintenance of traffic signals is essential for proper operation. Neglect can lead to danger to road users and additional costs to the public and the road authority responsible for the traffic signals.

2 This chapter provides an overview only of aspects related to the *management* of traffic signal maintenance. Technical procedures and methods of maintaining traffic signals are not discussed – such information can be obtained from the manufacturers of signal equipment. Some information in this regard can also be obtained from the references given in the bibliography.

3 The chapter is further restricted to the management of traffic signal components only, and does not cover other work such as the road pavement, road markings, road signs, etc. Remedying defects that occur during the contractual defects liability period and which are covered under the implementation contract, is also not covered in this chapter.

28.2 CONSEQUENCES OF MAINTENANCE DEFICIENCIES

1 The cost-effectiveness of good signal maintenance is beyond question. The traffic safety implications of malfunctioning traffic signals alone can justify the cost of keeping signals in proper working order.

2 The possibility of liability judgements against a road authority is also an important consideration in signal maintenance. Proper maintenance of traffic signals is of vital importance if liability is to be minimised.

3 It is particularly important that appropriate maintenance procedures should be in place AND strictly adhered to by personnel. These maintenance procedures must demonstrate a thoroughness, and must reflect a duty to care and to report defects. The maintenance procedure can be used as evidence regarding the standard of care taken by the road authority concerned.

4 It is also particularly important that proper maintenance records be kept so that in the event of litigation there will be no doubt, as to the nature and extent of the maintenance actions taken, and when such actions were taken. Many litigation actions would tend to shift the blame for an accident onto some aspect of the traffic signal.

28.3 REDUCING MAINTENANCE REQUIREMENTS

1 The maintenance of traffic signals can require a significant expenditure on manpower and material. An effort should therefore be made during the design and installation stages to implement systems that would contribute to minimising maintenance costs.

2 The following are a number of the factors that can contribute to high maintenance costs:

(a) Inappropriate selection of technology and highly sophisticated systems without due consideration of available skills and maintenance funds.

(b) Inadequate inspection during installation is often an important reason for maintenance problems. Some deficiencies are of such a nature that they cannot be detected, even with proper inspection. It is therefore important that responsible and experienced persons should undertake signal installations.

(c) The use of poor quality or non-standard equipment will require more skills and greater inventories of spare parts. Different types of equipment may require different maintenance procedures, which increase the management effort.

(d) Not providing adequately for maintenance in the design of traffic signal components.

3 An important factor in the design of signals is the provision for resistance against damage due to power surges, such as those resulting from lightning. Power surges can result in the destruction of all electronic components at a traffic signal, which will then have to be replaced at high cost.

28.4 TYPES OF MAINTENANCE

1 Maintenance falls into two broad categories: **routine** and **repair** maintenance. The term "maintenance" is generally taken to mean both these operations.

2 Routine maintenance consists of a periodic inspection and servicing of traffic signals.

3 Repair maintenance is undertaken in direct response to reported failures at traffic signals.

28.5 ROUTINE MAINTENANCE

28.5.1 General

1 Routine maintenance should primarily be aimed at those components where pro-active intervention is required. It is not cost-effective to service all signal components regularly while they are still in operating condition. It is more important that attention should be given during routine maintenance to the identification of signal components that are faulty.

2 Routine maintenance actions would normally include those recommended by the manufacturer of the traffic signal equipment. Different manufacturers may have diverse maintenance requirements for their equipment. Information on these requirements must be obtained from the manufacturer.

3 Other routine maintenance actions are summarised in Table 28.1. The actions listed in the table are discussed in following subsections.

4 The intervals at which routine maintenance actions are undertaken depend on a variety of factors, such as the availability of a remote monitoring system. One possible method is to implement a regular inspection system and to undertake routine maintenance only when required.

28.5.2 Routine controller cabinet maintenance

1 The maintenance of the controller cabinet involves checking of the cabinet door for proper operation and ensuring that the cabinet is not damaged or dirty. Weatherproof seals must be in a good condition to prevent moisture entering the cabinet. The cabinet must be kept clean, and particular attention must be given to possible insect infestations.

2 Infestations by insects, especially ants, and other pests can be a serious problem at traffic signal installations. Special insecticides should be used that do not damage the electrical and electronic components. Infestations are also indicative that the cabinet or enclosure is not adequately sealed against elements.

28.5.3 Routine signal head maintenance

1 The maintenance of signal heads includes activities such as lamp replacement, cleaning of lenses and reflectors, alignment of the signal heads and inspection for damage.

2 It is important that lenses should be fitted and sealed properly to prevent the ingress of moisture and dust. Ingress of moisture or excessive condensation within the signal head housing may cause oxidation of metal parts and problems with electrical connections and transformers for quartz-halogen lamps.

28.5.4 Routine signal post maintenance

1 Signal post maintenance covers aspects such as checking for accident and other damage as well as the repainting of the posts.

2 Posts can usually not be repaired when damaged in an accident, and are then replaced with new posts.

28.5.5 Routine push button maintenance

1 Push button maintenance is required to ensure that push buttons are operating properly and that they are not damaged.

2 The maintenance action may have to be extended to also include special devices provided for disabled pedestrians.

28.5.6 Routine detector loop maintenance

1 The vehicle detector loop is one of the more vulnerable components of a traffic signal system. They can be a frequent source of trouble if they not correctly installed. The normal movement of a flexible road pavement, together with the ingress of material into the loop slots, can cause abrasion with eventual breakdown of the insulation on the loop wires.

2 The reliability of detector loops can be significantly improved by following proper installation procedures. These procedures are described in Chapter 20 of this manual (Volume 3).

TABLE 28.1: ROUTINE MAINTENANCE

Maintenance task
Controller cabinet (including pedestal)
Door, hinges and locks
Weatherproof seal (gasket)
Damage, rust, dirt, insect infestation, etc.
Signal heads (lenses, reflectors, visors, background screen, etc.)
Clean signal heads (particularly lenses)
Signal head alignment
Damage to signal heads and lamps
Damage includes broken lenses, faded colours
Signal posts (including anchor bolts)
Repainting of posts
Damage and rust
Post support and anchors
Other road signs
Damage and rust
Anchoring of signs
Push buttons
Push button operation
Damage and rust
Loop detectors
Visual inspection of loops
Verify detector operation
Pull-boxes and manholes
Damage to covers and installations
Excessive water, dirt, insect infestation
Cables and other electrical
Cable joints (those that are visible)
Circuit breakers
Earthing
Controllers
Electromechanical assemblies
Relays, Flashers, Switches, Connectors
Lamp and fault monitor
Conflict monitor
Communication devices
Lightning protection unit (if testable)
Backup batteries (if provided)
Other as per manufacturer requirements

28.5.7 Routine maintenance of draw boxes and manholes

1 The maintenance of draw boxes and manholes involves checking them for damage and cleanliness. Checks should also be made for infestations by insects and other pests.

2 Theft of metal manhole covers may be a problem in certain areas. Consideration should then be given to using covers made of concrete or similar material.

28.5.8 Routine maintenance of electrical cables and other electrical components

1 Maintenance is required for all electronic components, including cables, cable connections, circuit breakers and earthing.

2 Cables can be vulnerable to damage during traffic accidents, and be a source of great danger when exposed. Maintenance efforts can be minimised if adequate provision is made for fail-safe designs or the easy replacement of cables.

28.5.9 Routine controller maintenance

1 The maintenance of a signal controller is a complex task and must be undertaken by skilled personnel. Routine maintenance may include tasks such as running diagnostic tests, replacing defective or damaged modules as well as printed circuit boards.

2 The advice and instructions of the manufacturer on maintenance and repair procedures, including testing and diagnostic procedures, should be closely followed. Where special tools or testing equipment are necessary, these should be made available to the maintenance and repair personnel and should be kept in good working condition.

3 Faults in electromechanical controllers can often readily be detected by visual inspection. Reasons for breakdowns include blown fuses, in-operative motors, contact points burnt or welded together, and camshaft lugs broken. Repair of electromechanical units is often relatively simple, provided spares are readily available.

4 Electronic controllers may be repaired by replacing defective circuit boards or modules. A controller will usually comprise several separate and independently replaceable boards or modules mounted on a motherboard or back plate.

5 Where a backup battery is provided in the controller, it is important that such battery be checked occasionally and replaced according to manufacturer's recommendations.

6 Where possible, lightning protection units should also be checked regularly and be replaced when defective.

28.5.10 Routine maintenance of road signs and road markings

1 The routine maintenance of road signs and markings provided at traffic signals is necessary when they are faded, damaged or not properly anchored to posts.

2 The maintenance of road signs and markings would usually not be undertaken by the signal maintenance team, and a separate team would normally be assigned for such work. It is, however, important that this team be informed of defects when they occur.

28.6 REPAIR MAINTENANCE

28.6.1 Introduction

1 Repair maintenance is aimed at repairing reported malfunctions in traffic signals. Such malfunctions occur due to reasons such as:

(a) Signal components reaching functional obsolescence.

(b) Poor construction techniques or substandard quality products.

(c) Traffic accidents, vandalism and natural causes, such as lightning, heavy winds, ingress of moisture, insects, etc.

(d) Environmental (dirt and dust).

2 Many malfunctions are caused by traffic accidents, but poor routine maintenance could also be contributing to a high incidence of signal faults.

28.6.2 Fault detection

1 Faults in signal operations can be detected in a number of ways. These include the following:

(a) Police reports of damage as a result of accidents.

(b) Complaints received from the public, the police or other departments of the road authority.

(c) Computer based detection of malfunctions.

(d) Routine inspections by a maintenance team.

2 Due to the high incidence of damage resulting from traffic accidents, standard procedures should be put in place according to which police can report such damage.

3 The public also plays an important role in reporting malfunctions. It is particularly important that an easy reporting procedure should be implemented which is widely publicised. A central fault reporting telephone number used for all types of municipal problems is a very effective method of allowing the public to report any faults.

4 The availability of a central computer (remote monitoring system) can simplify the detection of malfunctioning equipment significantly. A variety of diagnostic tests can be undertaken of the various electronic components in a signal system, and possible errors logged for further attention by a maintenance team.

5 The maintenance teams themselves can be utilised to undertake routine inspections when they are not attending to emergencies. These teams can undertake the routine repairs themselves, or work orders may be issued for more specialised repair work.

28.6.3 Response to malfunctions

1 Responding to a malfunction requires a formal system whereby a maintenance team can be instructed to undertake the required repair work.

2 Certain malfunctions require a more urgent response than others. Malfunctions that pose a public danger should receive priority.

3 Emergency malfunctions are those that create an immediate danger to the public or traffic, and should be attended to immediately. A 24-hour standby service should be available to handle such malfunctions. The following are examples of malfunctions that should be considered as emergencies:

(a) Any malfunction causing all red signal aspects failing on one or more approaches at a signalised junction.

(b) Displaced signal aspects resulting in the display of green signals to conflicting traffic streams.

(c) Signal posts and other components creating a hazard on the roadway.

(d) Exposed electrical cables that could result in life-threatening situations or in short-circuits.

(e) Opened or damaged controller cabinets resulting in exposed electronic or electrical components.

4 Other malfunctions that do not create immediate dangerous situations do not have to be treated as emergencies, and can be attended to at a more opportune time. Priority should, however, be given to those malfunctions that have the most disruptive effect on traffic flow, or which result in the highest levels of congestion.

5 The following response times, between the time that the maintenance team was informed of a malfunction and the time of arrival of the team at the site, are generally recommended:

(a) Emergency malfunctions that may have safety implications – The response time should not exceed two hours for 24 hours of the day, seven days of the week. Maintenance work undertaken in response to these malfunctions can be restricted to emergency repairs only.

(b) Major malfunctions that could seriously effect traffic flow, but not road safety – These types of malfunctions do not need immediate response, particularly if such malfunctions occur at night or over weekends. During weekdays, a response should be required on the same day if the malfunction is reported before 12:00. When reported after 12:00, or over the weekend or a public holiday, a response should be required before 12:00 on the next available workday.

(c) Minor malfunctions that would not seriously affect both traffic flow and road safety – A response time of longer than one day can be given, depending on the type of the malfunction.

6 Responding to malfunctions does not mean that a fault must be repaired immediately and at any cost. Maintenance work can be restricted to emergency repairs to make the signal safe, and the permanent repair work can be done at a more opportune time when manpower and materials are available for the work.

7 **A signal can be made safe by placing it in flashing mode, but where this is not possible the signals may be switched off. In cases where repair will take some time, consideration may be given to masking the signals out, and using temporary stop signs for the control of the junction or crossing.**

28.7 MAINTENANCE RECORDS

1 Detailed recording of fault reports and maintenance activities is an essential part of traffic signal maintenance. Such records are not only required for litigation purposes, but also form an important part of the management process. A file should be opened for each signal installation and record kept of each action taken at the installation.

2 An effective recording system is also essential where maintenance is contracted out and there are maximum time limits prescribed for action by the contractor, with commensurate penalties for failing to meet them.

3 Additional information on maintenance records is provided in Chapter 30 of this manual (Volume 3).

28.8 MAINTENANCE PERSONNEL

1 Required maintenance personnel levels and skills are discussed in Chapter 25 of this manual (Volume 3). A maintenance team should consist of qualified electricians, line workers and worker assistants.

2 Electricians are responsible for the maintenance of the electrical and electronic components of traffic signals. Line workers are responsible for tasks such as lamp replacement, cleaning of lenses, painting of posts and alignment of signals.

3 At least one emergency response team should always be on standby, 24 hours of the day, even if the team has limited capabilities. This team should be able to undertake emergency repairs.

4 IMPORTANT NOTE: All electrical work may only be carried out by a qualified electrician and under no circumstances should anyone else open a cabinet or enclosure containing electrical apparatus, live cables or exposed electrical terminals where there is a danger of electric shock.

28.9 MAINTENANCE RESOURCES

1 Adequate resources and spare parts should be available to allow maintenance personnel to undertake and complete maintenance work.

2 It is important that an administration facility be provided from where the maintenance activities can be managed and where malfunctions can be reported.

3 Teams should be provided with service vehicles equipped with all the required tools, spares and consumables required for the maintenance of traffic signals. Radios or cellular telephones should be provided for on-street communication. At least one truck must also be available that will allow personnel access to overhead signals.

4 Larger road authorities may also provide a workshop facility where signal components can be repaired. Smaller authorities would probably employ contractors for such maintenance duties.

28.10 CONTRACT MAINTENANCE

1 Traffic signal maintenance can readily be undertaken on a contractual basis. Smaller road authorities can probably not justify employing a full-time staff complement for maintaining signals. For larger road authorities, however, there could also be advantages in employing a contractor to undertake the signal maintenance.

2 Two types of contract maintenance can be considered. Firstly, the contract is between two separate road authorities with adjoining jurisdictions. Under this arrangement, the one road authority can maintain traffic signals on behalf of the other road authority. Payment for the service is usually a flat annual amount.

3 The second type of maintenance contract is with a private contractor. The contract should normally include only those tasks that are well defined, such as group replacement of signal lamps, painting of posts, etc. Where new installations are constructed by private contractors, a period of contract maintenance may be a part of the installation contract.

4 Provision should be made in the contract for two types of maintenance. The one type relates to routine inspections and maintenance, including the repair of defects other than those caused by damage. The other type will cover the repair of damage resulting from collision, explosion, earthquake, flood, riot and other incidents.

5 An essential element in any maintenance contract is the establishment of an effective system for recording the details of defects and damage as they are reported, and of subsequent repair work when it is completed. The records should clearly indicate the category of work, for payment purposes. Dates and times of these events should always be carefully recorded.

6 To ensure timely performance by the contractor, it is important to specify maximum time limits for the response to a reported malfunction, and for effecting repairs (see Section 28.6.3). If the contractor fails to meet a stipulated deadline for a single call-out, a penalty should be specified, usually in the form of a rebate to the road authority, per hour or other period by which he exceeds the time limit. The response times can be negotiated with contractors with the aim of reducing costs, as long as road safety is not compromised.

28.11 BIBLIOGRAPHY

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